Morphology-based phylogenetic analysis: taxon sampling, character formulation and character coding

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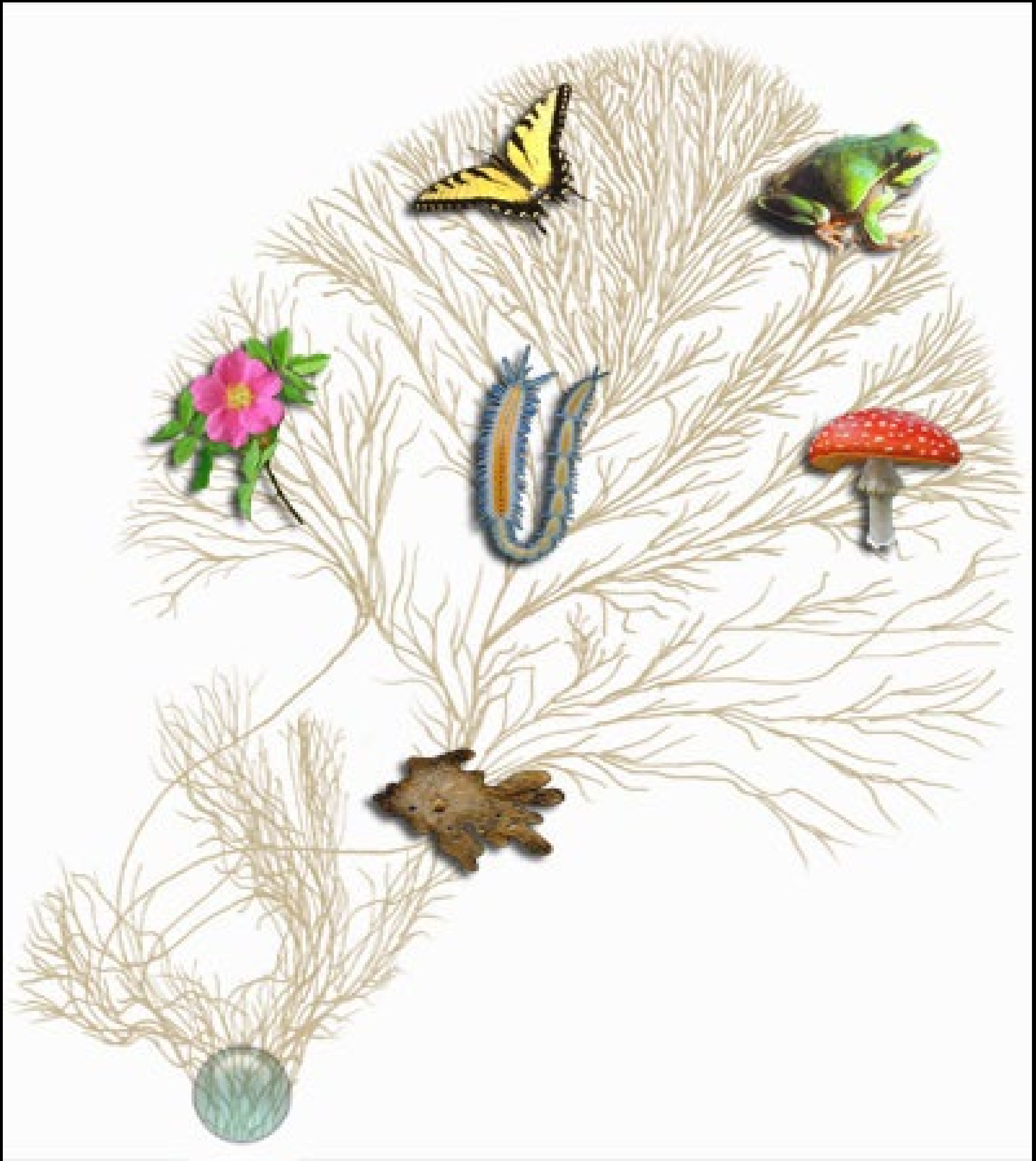


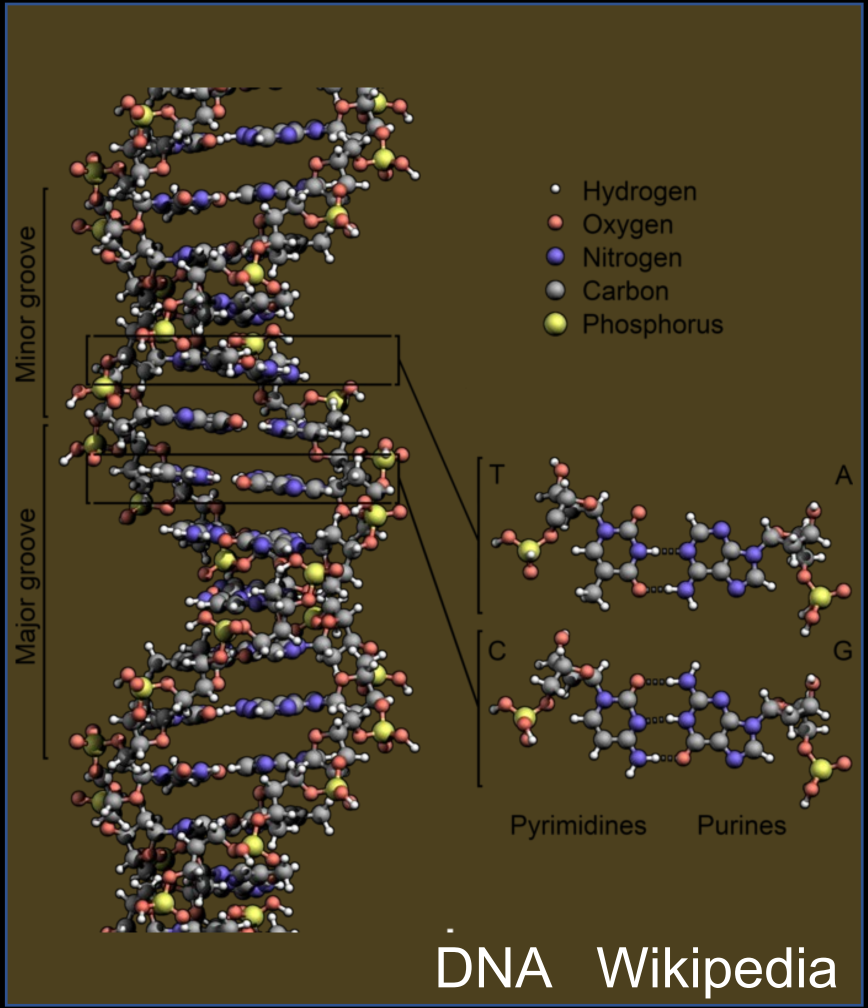
Tree of Life

http://tolweb.org/tree

Two types of data

Phenotypic data:
morphology, development, physiology, and behavior

Molecular data:
DNA and protein sequence



性状-分类单元矩阵（Character-Taxon matrix）

- Taxon (T)
- Character (C)
- Matrix

	C1	C2	C3
T1	A	G	T
T2	A	G	T
T3	A	C	T
T4	A	C	A

	C1	C2	C3
T1	0	0	0
T2	1	0	1
T3	1	1	1
T4	2	1	1

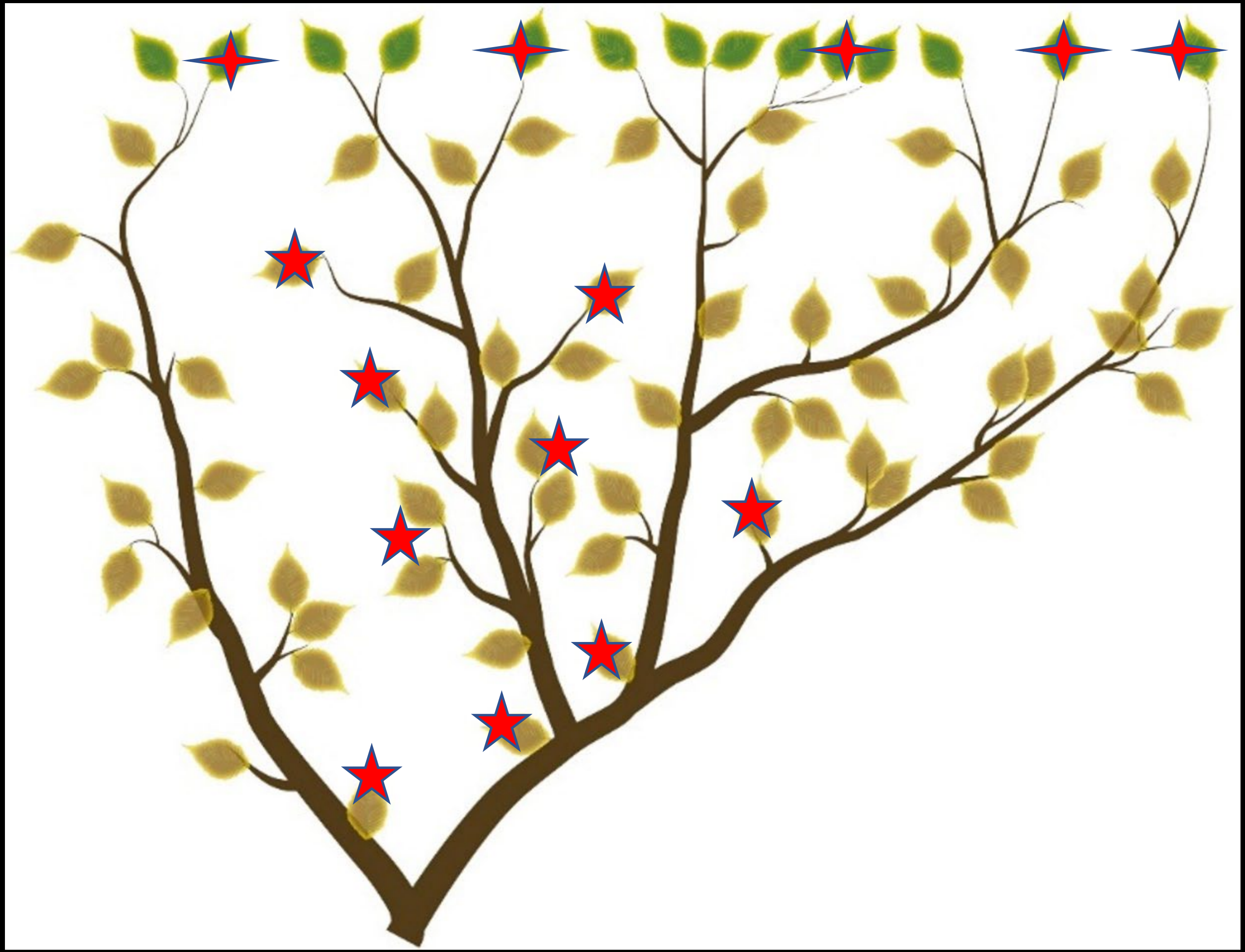
Different methods

- Distance-based methods
- Parsimony-based methods
- Likelihood-based methods (Maximum likelihood)
- Posterior-probability-based methods (Bayes)

Morphology/phenotype-based phylogenetic analysis

- Taxa (limited in number but covering a wide temporal range)
- Characters (limited in number but covering different data)
- Methods (Parsimony and others)

Taxon sampling: terms and data carriers



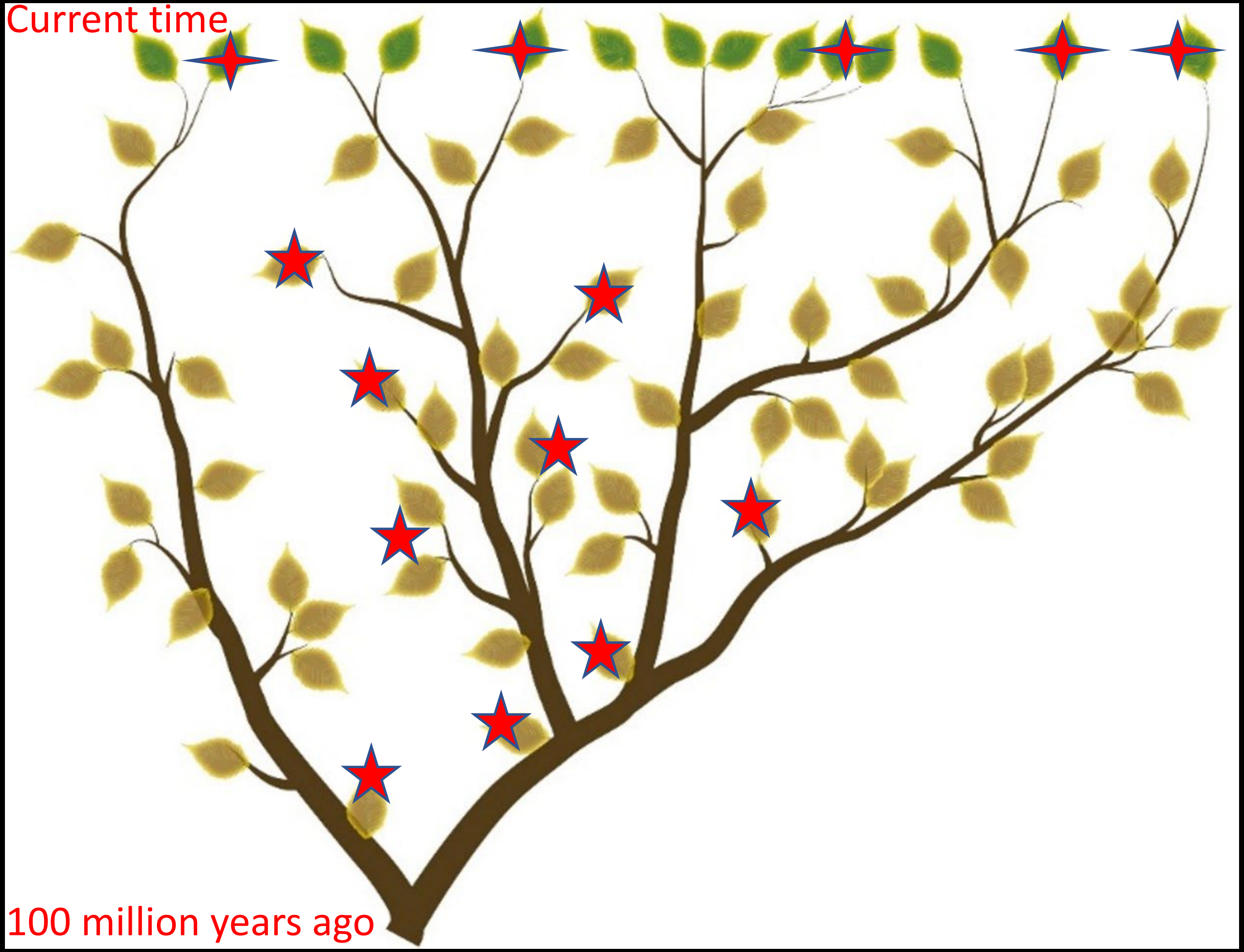
A few terms

- Taxon/Terminal Taxon/Tip (leaf)
- Operational taxonomical units (OTUs)

Data carriers

- OUT = species/genus = specimen (e.g., organs)

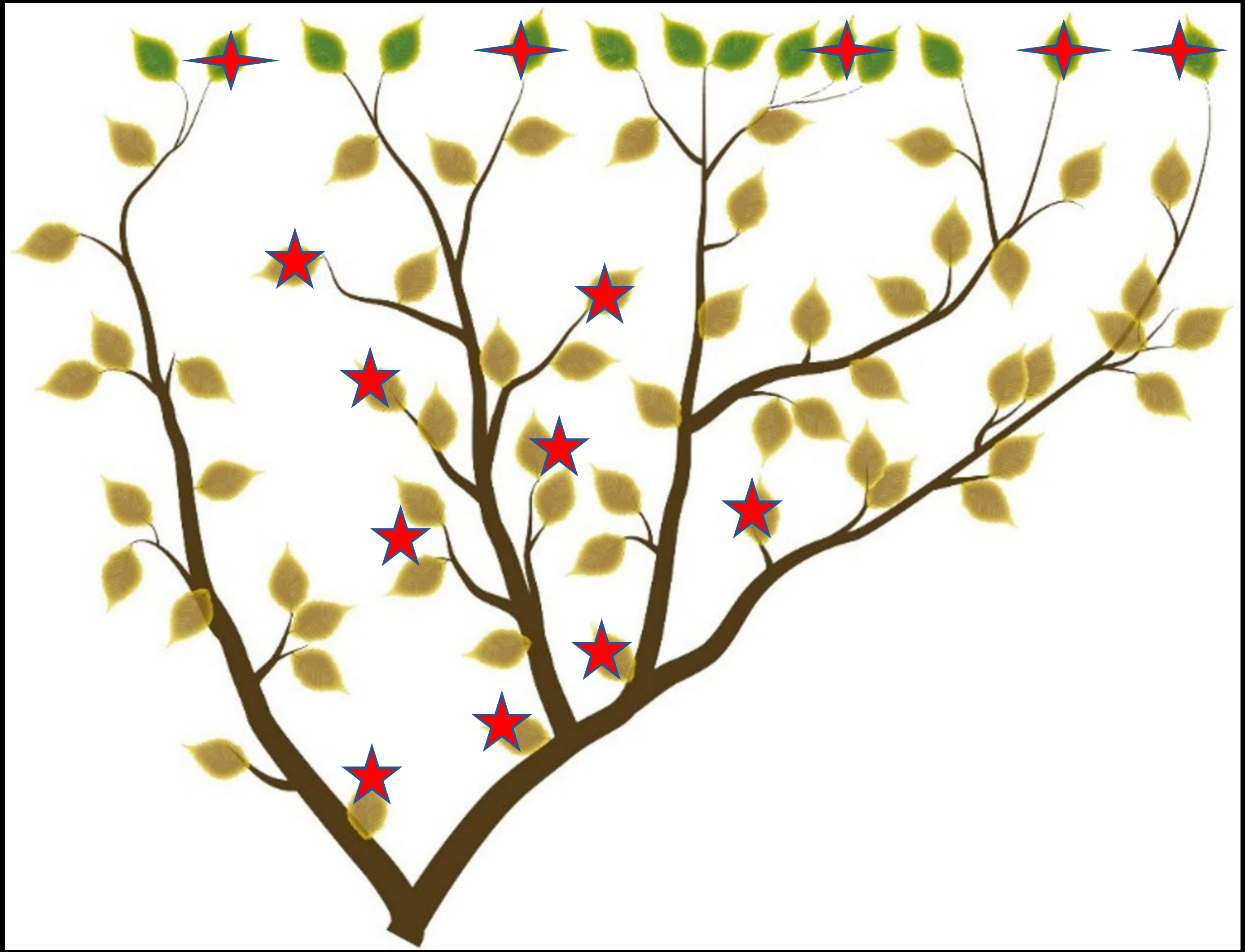
How to select taxa for the analysis?



- Limited taxon sampling
- Sampling principles

1. Covering major lineages
2. Typical species
3. Early-branching species
4. > 1 outgroup

Taxon sampling size?



- More taxa added in, more robust the results
- Platform effect of increasing taxon samples

Phenotypic dataset construction

Phenotypic character: a set of comparable phenotypic variations (the same but different)

1. Toe number: 5 (0); 3 (1); 1 (2)
2. Somatic maturity age: > 1 year old (0); < 1 year old (1)
3. Thermoregulation: ectothermy (0); endotherm (1)
4. Brooding behavior: absent (0); present (1)

	C1	C2	C3	C3
S1	0	0	0	0
S2	1	0	1	1
S3	1	1	1	0
S4	2	1	1	1

Characteristics of Phenotypic Dataset

- Weakness: fewer characters (< a few thousands), high time cost and subjectivity in data collecting
- Strength: covering both living and extinct species, ?strong phylogenetic signals
- Neutral: polymorphism rare

Formulation of morphological characters

Sereno 2007 Cladistics:

- Characters: features expressed as independent variables
- Character states: the mutually exclusive conditions of a character
- Character types: neomorphic and transformational
- Character statements: containing four fundamental functional components, i.e., locator, variable, variable qualifier, and character state

Examples：

locator variable variable qualifier character state

1. Humerus, length relative to femur: <50% (0) or > 50% (1) (transformational)
2. Frontal, bony horn: absent (0) or present (1) (neomorphic)

Characters are independent variables, logically but not biologically

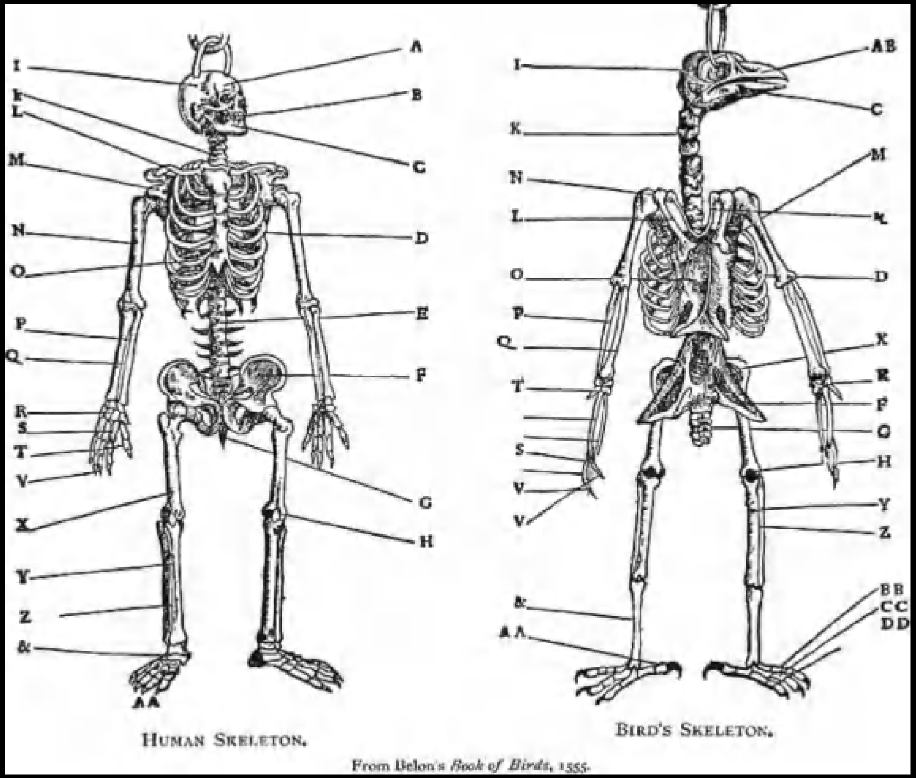
1. Coracoidal contribution to glenoid fossa: small (0); large (1)
2. Scapular contribution to glenoid fossa: small (0); large (1)

1. Dentition: blunt (0); sharp (1)
2. Dentition, denticles: coarse (0); fine (1)
3. Manual claws: blunt (0); sharp (1)

Formulation of morphological characters:
Primary homology assessment

Criteria for primary homology

- Topology and connectivity (rod-like wrist bones in crocodiles)
- Shape (dinosaur feathers)
- Transitional form in ontogeny and/or evolution (embryonic origin of mammalian ear bones)



Belon 1555

Characters for morphology-based phylogenetic analysis

- Binary character and multi-state character
- Synapomorphy (Phylogenetically informative characters)
- Autapomorphy, apomorphy, and plesiomorphy

	C1	C2	C3
S1	0	0	0
S2	1	0	1
S3	1	1	1
S4	2	1	1

	C1	C2	C3
S1	0	0	0
S2	0	0	1
S3	0	1	1
S4	1	1	1

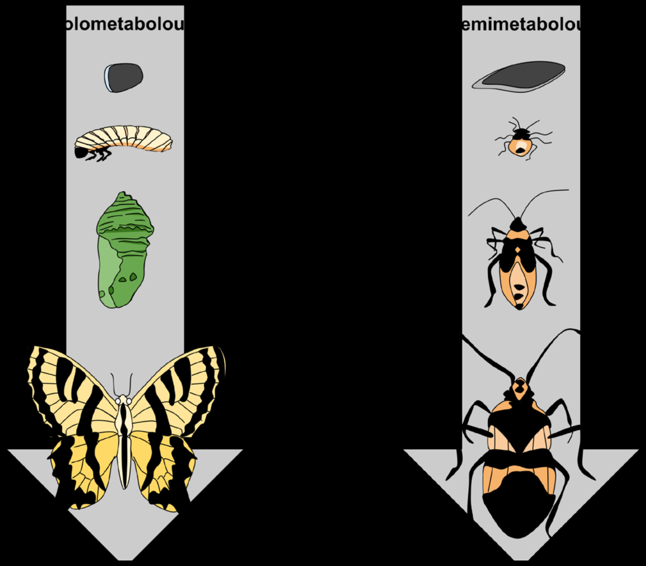
How many characters are enough?

- The more, the better (but platform effect)
- Pending on the question
- Cost

Issues in character formulation:
various intraspecific variations

Individual/population variations

Dimorphism

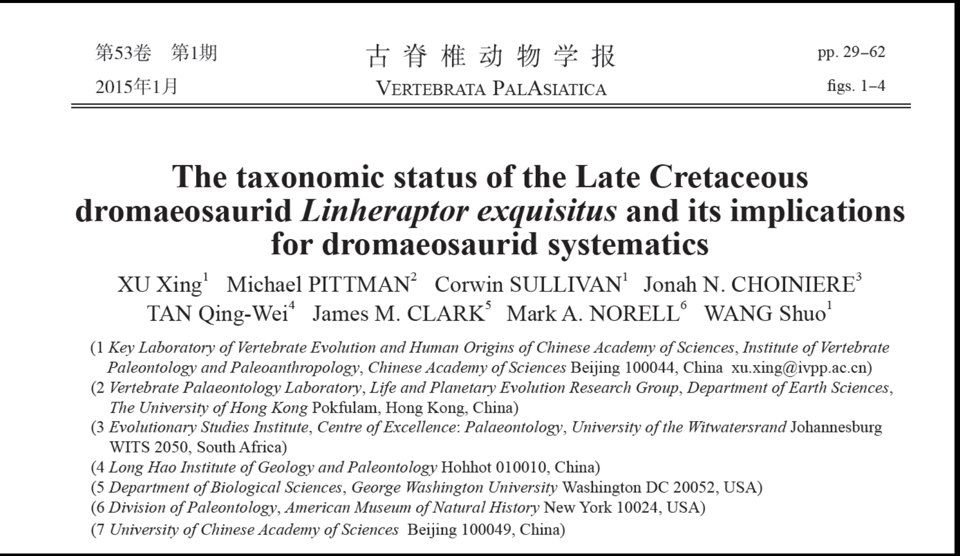


Ontogenetic variations

Wikipedia

Xu et al, 2015:
Different from other closely related species in 61 morphological features, but are they taxonomical differences or caused by ontogeny, dimorphism, and individual variations?

Lingheraptor exquisitus
A junior synonym of *Tsaagan mangas*
Turner et al, 2012
Senter et al, 2012
Evans et al, 2013



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The taxonomic status of the Late Cretaceous dromaeosaurid *Linheraptor exquisitus* and its implications for dromaeosaurid systematics

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Tsaagan mangas

Lingheraptor exquisitus

Velociraptor mongoliensis

Issues in character formulation:
combining transformational and neomorphic states

Example

1. Dentition: present, numerous (0); present, few (1); absent (2)

$\Downarrow$

1. Dentition: present (0); absent (1)
2. Dentition: numerous (0); few (1)

Issues in character formulation

Composite characters

Examples

1. Forelimb: short (0); long (1)

1. Humerous: short (0); long (1)
2. Radius : short (0); long (1)
3. Metacarpals : short (0); long (1)
4. Fingers : short (0); long (1)

1. Dentition: cone-shaped, without denticles (0); leaf-shaped, with coarse denticles (1); knife-shaped, with fine denticles (2)

1. Dentition: cone-shaped (0); leaf-shaped (1); knife-shaped (2)
2. Dentition, denticles: absent (0); present (1)
3. Dentition, denticles: coarse (0); fine (1)

Issues in character formulation:
poorly or impropterly defined states

1. Hand: short (0); long (1)
2. Dentition: numerous (0); few (1)
3. Coracoidal tubercle: small (0); prominent (1)
4. Humeral head: highly reduced (0); well developed (1)

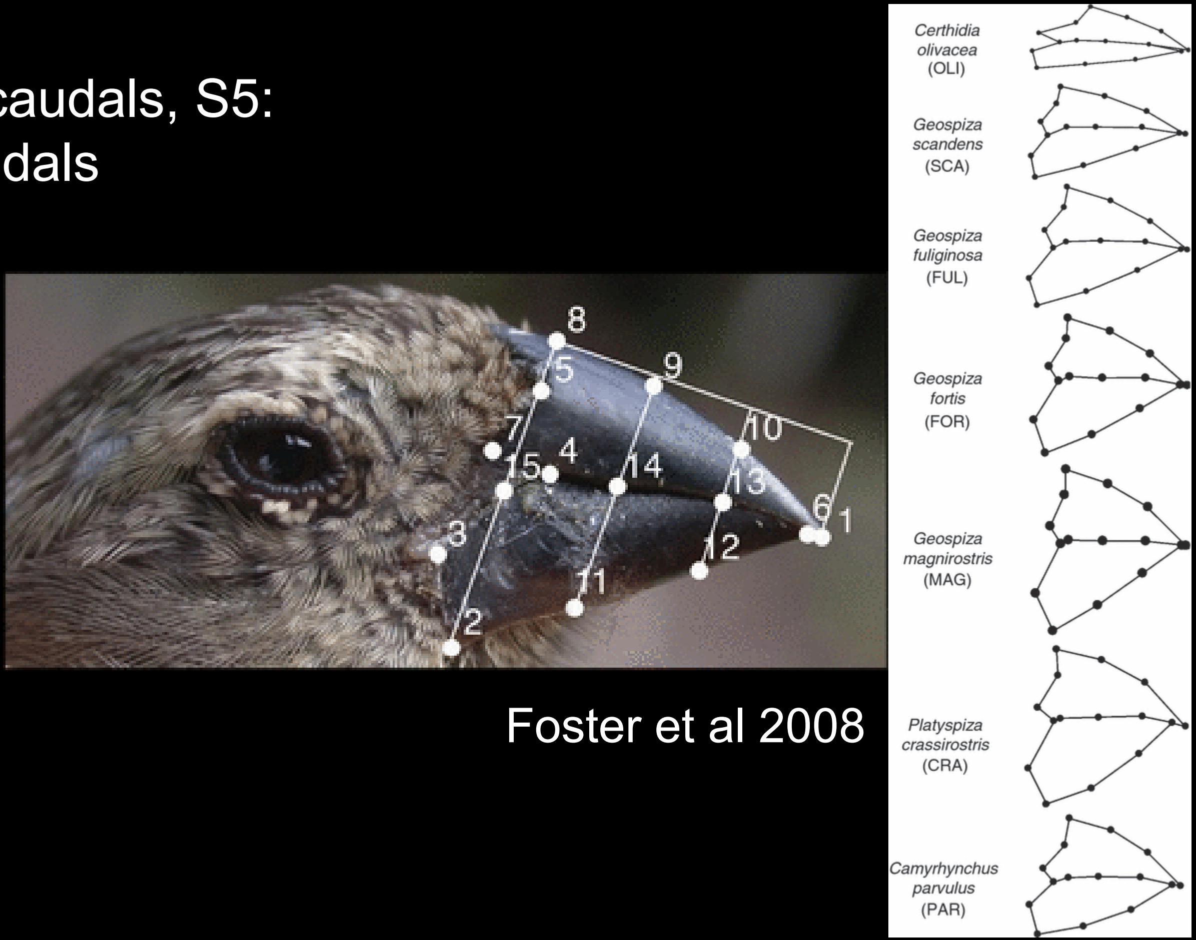
Non-discrete characters: numbers and shapes

- Continuous characters

S1: 5 caudals, S2: 10 caudals, S3: 30 caudals, S4: 31 caudals, S5: 35 caudals, S6: 39 caudals, S7: 60 caudals, S8: 63 caudals

Caudal number: >50 (0); 50-20 (1); <20 (2)

- Geometric morphometric characters



Foster et al 2008

- How to deal with features pertaining to proportional change: gap or segment-coding; statistical analyses; TNT

Ordering multi-state characters or not?

- Unordered and ordered
- Objective or subjective ordering?
- Matrix/Sankoff characters

1. Frontal horn cross section：circular (0); rectangular (1); triangular (2)
2. Frontal horn size: small (0); medium (1); large (2)

Character weighting

1. Feathers: absent (0); present (1)
2. Humerus, length relative to femoral length: shorter (0); longer (1)

- Homoplasy and character weighting
- Implied weighing (Goloboff 1993)

Phylogenetic reconstruction based on paleontological data

- Fossils: with chronological information (but with uncertainties)
- Fossilization and taphonomy: missing data (incomplete specimen issue, soft tissue issue, and species or genus as OUTs?); data quality (fossil deformation)
- Stratocladistics (地层分支系统学， Fisher 1994): StrataPhy, Marcot and Fox 2008; Bayesian tip-dated methods (贝叶斯末端定年方法， Ronquist et al. 2012)

Morphology-based Phylogenetic analysis: solutions

- MorphoBank
- Character illustration
- Character ID
- Automation (Artificial Intelligence)

Thanks！

Questions？